COMSATS University Islamabad, Lahore Campus
Department of Electrical and Computer Engineering

Assignment 2 – SPRING 2026

Course Title:	Fundamentals of Digital Logic Design			Course Code:	EEE240	Credit Hours:	3(2,1)
Course Instructor:	Dr. Ali Raza			Programme Name:	BSE		
Semester:	3rd	SP25	Section:	A	Date:	12-05-26	
Submission Date:	14th April, 2025			**Maximum Marks:**		40	
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Important Instructions / Guidelines:

- In case of plagiarism, you will be marked ZERO.

Problem 1:

Apply gate minimization technique using 4-variable K-map to simplify the following Boolean expression:

$$\bar{A}\bar{B}\bar{C}\bar{D} + A\bar{C}\bar{D} + \bar{B}C\bar{D} + \bar{A}BCD + B\bar{C}D$$ (Standard SOP)

$\overline{ABCD} + A\overline{CD}B + A\overline{CD}\bar{B} + \bar{A}\bar{B}C\bar{D} + \bar{A}BC\bar{D} + \bar{A}BCD + B\bar{C}DA + B\bar{C}D\bar{A}$

Simplified expression: $\bar{A}BD + AB\bar{C} + \overline{BD}$ (K-map on back page)

Problem 2:

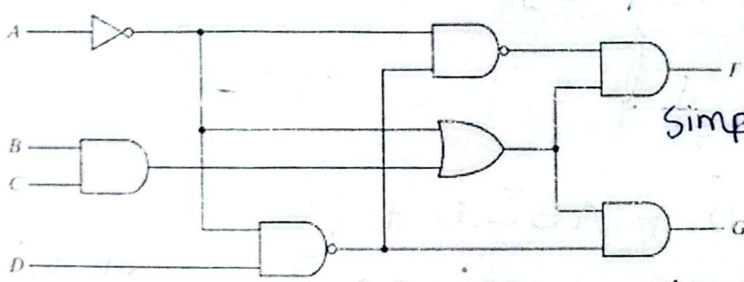
Analyze the following Boolean function F, together with the don't-care conditions d, to express function in sum-of-minterms form:

$$F(A,B,C,D) = (0,6,8,13,14)$$
$$d(A,B,C,D) = (2,4,10)$$

Simplified expression: $\bar{B}\bar{D} + C\bar{D} + AB\bar{C}D$

Problem 3:

Using digital circuit analysis technique, evaluate the Boolean expression for output F and G in terms of the input variables A, B, C and D. Then use K-map to simplify the expression.



$F = \overline{\bar{A}.\overline{\bar{A}.D}} . \bar{A} + B.C$

Simplified: $(A+D)$

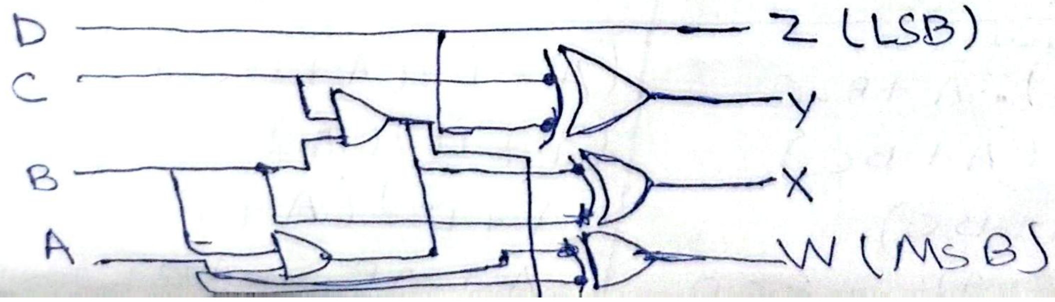
$G = \overline{\bar{A}.D} . \bar{A} + B.C$

Simplified: $(A+\bar{D})$

Problem 4:

Design a 4-bit combinational circuit that generates the 2's complement of a 4-bit binary input number. Provide the truth table, Karnaugh map (K-map), and the resulting circuit diagram in your working.

Inputs Output



Problem#1: Kmap: $F = \Sigma($

AB \ CD

AB \ CD	00	01	11	10
00	1	0	0	1
01	0	1	1	0
11	1	1	0	0
10	1	0	0	1

$= \bar{A}BD + AB\bar{C} + \bar{B}\bar{D}$

Problem#2: Kmap

AB \ CD	00	01	11	10
	1	0	0	X
	X	0	0	1
	0	1	0	1
	1	0	0	X

$= \bar{B}\bar{D} + AB\bar{C}D + C\bar{D}$

Problem#3: Expression: $(\overline{\bar{A}\cdot\overline{(\bar{A}\cdot D)}})(\bar{A}+B\cdot C) = F$

$(\overline{\bar{A}\cdot D})(\bar{A}+B\cdot C) = G$

F	G
de'morgan's law: $(\bar{\bar{A}}) + (\overline{\bar{A}\cdot D}) \cdot \bar{A} + B\cdot C$	$(\bar{\bar{A}} + \bar{D})(\bar{A} + BC)$
$(A + \bar{A}\cdot D)(\bar{A} + BC)$	$(A + \bar{D})(\bar{A} + BC)$
$(A + D)(\bar{A} + BC)$	$(A + \bar{D})(\bar{A} + BC)$
$A + ABC + \bar{A}D + BCD$	$A + ABC + A\bar{D} + BC\bar{D}$

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K-map:

F

AB \ CD	00	01	11	10
00	0	1	1	0
01	0	1	1	0
11	1	1	1	1
10	1	1	1	1

$= (A+D)$

G

CD \ AB	00	01	11	10
00	1	0	0	ϕ
01	1	0	0	ϕ
11	1	1	1	1
10	1	1	1	1

$= (A+\bar{D})$

Q4) Kmap for output Y:

CD \ AB	00	01	11	10
00	0	1	0	1
01	0	1	0	1
ϕ1	0	1	0	1
10	0	1	0	1

Kmap for output Z

CD \ AB	00	01	11	10
00	0	1	1	0
01	0	1	1	0
11	0	1	1	0
10	0	1	1	0

AB $Y = C \oplus D$. AB $Z = D$

Now

$W = A \oplus (B+C+D)$

$X = B \oplus C + D$

$Y = C \oplus D$

$Z = D$

Binary A	B	C	D	1's complement	Adding 1	2's complement W	X	Y	Z
0	0	0	0	1111	+ 1	0	0	0	0
0	0	0	1	1110	+ 1	1	1	1	1
0	0	1	0	1101	+ 1	1	1	1	0
0	0	1	1	1100	+ 1	1	1	0	1
0	1	0	0	1011	+ 1	1	1	0	0
0	1	0	1	1010	+ 1	1	0	1	1
0	1	1	0	1001	+ 1	1	0	1	0
0	1	1	1	1000	+ 1	1	0	0	1
1	0	0	0	0111	+ 1	1	0	0	0
1	0	0	1	0110	+ 1	0	1	1	1
1	0	1	0	0101	+ 1	0	1	1	0
1	0	1	1	0100	+ 1	0	1	0	1
1	1	0	0	0011	+ 1	0	1	0	0
1	1	0	1	0010	+ 1	0	0	1	1
1	1	1	0	0001	+ 1	0	0	1	0
1	1	1	1	0000	+ 1	0	0	0	1

K-map for Output Bit W (MSB)

AB \ CD	00	01	10	11
00	0	1	1	1
01	1	1	1	1
11	0	0	0	0
10	1	0	0	0

$W = A \oplus (B + C + D)$

K-Map for output X

AB \ CD	00	01	11	10
00	0	1	1	1
01	1	0	0	0
11	1	0	0	0
10	0	1	1	1

$X = B \oplus C + D$